

KALEEL RAHMAN MOHAMED SATHAM

Software Engineer | AI/ML Developer | Full-Stack Developer

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PROFESSIONAL SUMMARY

Motivated Software Engineer with project-based experience in AI/ML, Full-Stack Web Development, IoT Systems, and Enterprise Java Applications. Proficient in Java, Python, and JavaScript with hands-on expertise in React.js, Node.js, Express.js, and MySQL database integration. Demonstrated ability to design, develop, and deploy end-to-end software solutions including machine learning models (CNN, NLP, YOLOv8), RESTful web applications, and real-time IoT systems. Committed to writing clean, maintainable code and contributing effectively to Agile software development teams in an entry-level Software Engineering role.

TECHNICAL SKILLS

Languages	Java, Python, JavaScript
Frontend	HTML5, CSS3, React.js, React Hooks, JSP, Responsive UI Design
Backend	Node.js, Express.js, Java EE, Servlets, REST API, RESTful Web Services, MVC Architecture
Databases	MySQL, Microsoft SQL Server, Database Design, SQL Queries
AI / ML	Machine Learning, Deep Learning, CNN, NLP, Scikit-learn, YOLOv8, Object Detection, Image Classification, Text Classification, Transfer Learning, Feature Engineering, Model Training & Evaluation, Hyperparameter Tuning, Data Preprocessing
IoT	Arduino, ESP32, Sensor Integration, Real-Time Systems, Alert Systems
Tools	Git, GitHub, VS Code, IntelliJ IDEA, Eclipse, Figma, SDLC, Agile Methodology
Design	UI/UX Fundamentals, Wireframing, Prototyping, Technical Documentation

PROJECTS

Hate Speech Detection System | Python · Machine Learning · NLP · Scikit-learn

- Engineered an end-to-end NLP pipeline for automated hate speech classification, including text preprocessing, tokenization, feature extraction, and model inference.
- Designed and trained a supervised ML classification model on large-scale text datasets using Scikit-learn, achieving high accuracy through iterative hyperparameter tuning.
- Implemented data cleaning and normalization workflows to handle noisy, real-world text data, improving overall model robustness and evaluation scores.

Harmful Religious Image Detection | Python · YOLOv8 · Computer Vision · Deep Learning

- Architected a real-time object detection system using YOLOv8, managing the complete ML pipeline from dataset annotation and augmentation to model training and evaluation.
- Applied transfer learning techniques on a custom-annotated dataset to improve detection precision and reduce false positive rates.
- Conducted systematic performance optimization and model evaluation, achieving measurable improvements in detection accuracy across test sets.

Brain Tumor Detection System | Python · CNN · Deep Learning · Medical Imaging

- Developed a Convolutional Neural Network (CNN) model for multi-class brain tumor classification from MRI scan datasets using deep learning techniques.
- Preprocessed and augmented medical image datasets to improve model generalization, minimize overfitting, and enhance classification reliability.
- Performed hyperparameter tuning and model evaluation using standard metrics (accuracy, precision, recall), achieving consistent performance improvements across training cycles.

Ocean View Resort Reservation System | Java EE · JSP · Servlets · MySQL · HTML5 · CSS3 · JavaScript [[GitHub](#)]

- Built a full-stack Java EE web application for hotel reservation management using MVC architecture with JSP, Servlets, and MySQL backend integration.

- Implemented dynamic reservation booking, room availability management, and an admin dashboard — version-controlled throughout using Git and GitHub.
- Designed and delivered a responsive, user-friendly UI using HTML5, CSS3, and JavaScript, improving overall user experience and system usability.

Hospital Management System | *HTML5 · CSS3 · JavaScript · MySQL*

- Developed a full-featured web-based hospital management system with modules for appointment scheduling, patient records, doctor management, and billing.
- Designed a normalized relational MySQL database schema to support multi-entity hospital operations with efficient query performance.
- Implemented a clean, responsive front-end interface using HTML5, CSS3, and JavaScript, ensuring cross-browser compatibility and intuitive navigation.

Billing System | *Java · MySQL · OOP*

- Developed a desktop-based billing and invoice management application using Java and MySQL, applying Object-Oriented Programming (OOP) principles for modular, maintainable code.
- Implemented customer and product management modules with full CRUD operations and database integration, supporting end-to-end billing workflows.
- Performed system testing and validation to ensure data integrity, accurate invoice generation, and reliable application performance.

Fire Detection IoT System | *IoT · Arduino · ESP32 · Python · Sensor Integration*

- Engineered a real-time IoT fire detection system using Arduino/ESP32 microcontrollers with multi-sensor integration and an automated alert notification mechanism.
- Integrated temperature and smoke sensors with embedded firmware logic to ensure accurate and reliable fire event detection under varying conditions.
- Validated system performance through rigorous real-time testing, achieving sub-second response time for fire detection and alert triggering.

EDUCATION

BSc (Hons) in Software Engineering ICBT Campus, Colombo, Sri Lanka Currently Pursuing	<i>Expected: 2027</i>
Higher National Diploma (HND) in Computing – Software Engineering BCAS Campus, Sri Lanka	<i>May 2025</i>
Diploma in Information Technology BCAS Campus, Sri Lanka	<i>June 2023</i>

EXPERIENCE

Data Entry Operator ▸ Maintained high data accuracy and integrity through systematic data entry, validation, and quality assurance processes across large datasets. ▸ Developed strong attention to detail and time management skills, consistently meeting deadlines in a fast-paced work environment.	<i>Part-Time Sri Lanka</i>
PC Clinic Support Assistant ▸ Diagnosed and resolved hardware and software issues for users, gaining practical IT troubleshooting, customer support, and technical communication experience. ▸ Assisted in PC setup, OS installation, and peripheral configuration, strengthening foundational system administration and problem-solving skills.	<i>Volunteer Sri Lanka</i>

CORE COMPETENCIES

Problem Solving · Analytical Thinking · Teamwork & Collaboration · Communication · Time Management · Adaptability · Attention to Detail · Technical Documentation · Self-Learning · Agile SDLC · Version Control (Git)